

LT2326: Assignment 1 - Multilingual Tokenization and Language Modeling

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Abstract

I trained three identical small Transformer language models on a balanced English/Turkish/Chinese Wikipedia corpus, one per tokenizer: a character tokenizer, a shared byte-level BPE with 2,000 tokens (BPE-2k) and a shared byte-level BPE with 10,000 tokens (BPE-10k). The models are compared with bits per character (BPC) on the test set. BPE-10k is best in every language (overall 3.410 BPC vs. 3.589 for characters). BPE-2k ties with characters overall (3.593), but only because two opposite effects cancel: it helps English and Turkish and hurts Chinese, where many characters are cut into byte fragments. Analysing which language “owns” which vocabulary entries suggests why: Chinese receives about half of the shared vocabulary but needs about twice as many different tokens as the other languages to cover its text. Code: [GitHub link](#)

1 Summary of Main Findings

1. **Token counts:** Chinese needs by far the most tokens. Under BPE-2k it needs 410k tokens for 368k characters, i.e. more than one token per character, and 35.4% of its tokens are fragments of characters.
2. **Larger vocabulary:** Going from 2k to 10k reduces the number of tokens by 28% to 32% in all languages. In BPC the largest gain is for Chinese (−0.389, −5.9%) mostly because BPE-2k had hurt it.
3. **Best model.** BPE-10k has the lowest test BPC in English (1.972), Turkish (2.033) and Chinese (6.186) although it made fewer updates than the character model and has about the same size. My prediction that BPE-2k would be best was wrong.
4. **The overall tie hides opposite effects:** Character and BPE-2k tie overall (3.589 vs. 3.593)

Split	en	tr	zh
train	36,004	30,379	75,962
valid	4,540	3,735	9,554
test	4,472	3,792	9,430

Table 1: Number of sentences per split and language.

but BPE-2k is better for English and Turkish and worse for Chinese.

5. **Sequence length and BPC:** Within every language, the order of the tokenizers by tokens per 100 characters equals their order by BPC. This is an association; it is confounded with context coverage and model size.
6. **Vocabulary allocation:** Chinese gets 54% (2k) and 49% (10k) of the entries but needs about twice as many different tokens to cover its text. English and Turkish share only 16% to 18% of the entries, yet these short Latin pieces cover 70% and 60% of their text at 2k.
7. **Error analysis:** Sentences with rare Chinese characters are much more expensive under BPE-2k than under the character model (up to +3.9 bits per character). BPE-10k wins on sentences made of common words; BPE-2k wins on names and unusual strings.

2 Setup

2.1 Data

The data are sentence files of one sentence per line, with about 2.95M characters per language in `train/` and 368k in each of `valid/` and `test/` (Table 1). I used *all* of the training data (no subsampling) so every model sees exactly the same sentences. The training data were used to build the tokenizers and train the models, the validation data to monitor training and choose the best checkpoint, and the test data only for the final evaluation.

Two properties of the data matter for the interpretation. According to `metadata.json`, the English file is *Simple English Wikipedia*, whereas Turkish and Chinese are ordinary Wikipedia. The Chinese file mixes traditional and simplified characters (for example both 一个 and 一個 are among the first BPE merges). Because the corpus is balanced by characters, Chinese has many more (shorter) sentences.

2.2 Tokenizers

- **Character-level:** The vocabulary is every distinct character in the training text (9,442 characters plus `<unk>`, 9,443 in total). A character that appears only in validation or test is mapped to `<unk>`. This is rare: 0.013% of the validation tokens in English, 0.004% in Turkish and 0.070% in Chinese.
- **BPE (small and large):** Byte-level BPE from the Hugging Face tokenizers library,¹ trained on `tokenizer/balanced.txt` (its per-language character counts equal those of `train/` in `metadata.json`, so no validation or test text is used). There is one shared vocabulary for the three languages. I chose **2,000** and **10,000** tokens. With 2,000, only about 1,750 merges remain after the 256 byte symbols and the special token, so the tokenizer stays close to bytes; 10,000 is five times larger, large enough for many whole words and frequent Chinese characters, while the embedding table remains manageable for a small Transformer. I chose *byte-level* BPE because every text can be encoded without unknown tokens, and because a character-based BPE with 2,000 tokens could not even hold all Chinese characters. The price is that a rare Chinese character (3 bytes) can be split into pieces.

2.3 Models

All three models have the same architecture (Table 2). Only the vocabulary, and with it the input and output matrices, differs. Every vocabulary contains one extra end-of-sentence (EOS) token that is appended after each sentence. The training sentences of all three languages are shuffled once with a fixed seed and concatenated into one stream,

¹<https://github.com/huggingface/tokenizers>

Setting	Value
Type	decoder-only Transformer (causal self-attention), pre-LayerNorm, GELU
Layers / heads	2 / 4
Hidden / feed-forward size	256 / 1024
Context length	256 tokens, learned position embeddings
Dropout	0.1
Input / output embeddings	separate (no weight tying)
Optimiser	AdamW, learning rate 10^{-3} , weight decay 0.01, 200 warm-up steps, cosine decay to 10%, gradient clipping 1.0
Batch	64 random windows of 256 tokens, fp32
Seed	2326

Table 2: Architecture and hyperparameters (identical for all three models).

Model	Train tokens	Steps/ep.	Steps	Time
char	8,980,877	548	5,480	559 s
BPE-2k	6,072,092	370	3,700	275 s
BPE-10k	6,182,877	377	3,770	300 s

Table 3: Training tokens, updates and wall-clock time for the 10-epoch policy.

so each batch mixes the languages. The models are trained from scratch (no pretrained models) on one GPU (GTX 1080 Ti on `mltgpu`, PyTorch 2.4.1).

2.4 Training-control policy

Policy: every model gets exactly 10 complete passes (epochs) over the same training text. I chose this because the models then see the same underlying data the same number of times, which is easy to explain and to reproduce. Table 3 shows what this means in tokens and updates.

3 Results

3.1 Tokenizer statistics

The token counts show a clear difference between the languages. The character tokenizer produces one token per character by construction. BPE-2k is close to byte-level tokenization for Chinese, so a single Chinese character can require several tokens. BPE-10k has enough merges to represent many frequent Chinese characters and common words as single tokens.

3.2 Language-model results

The final test BPC results are shown below.

BPE-10k gives the best result in all three languages. BPE-2k improves over the character model for English and Turkish but is worse for Chi-

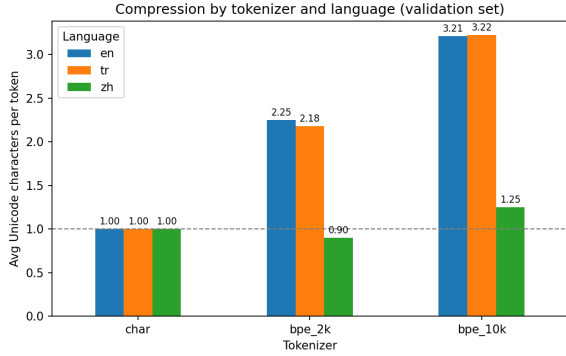


Figure 1: Characters per token on the validation set.

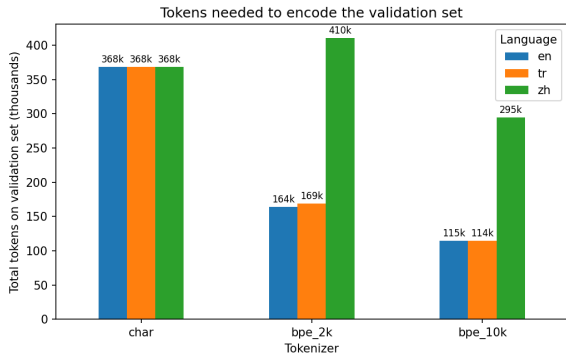


Figure 2: Total validation tokens by language and tokenizer.

nese. The overall result therefore hides different effects across languages.

3.3 Learning curves

Figure 3 shows the validation loss during training.

The character model receives more updates because its tokenized training stream is longer. This is a consequence of the fixed-epoch training policy rather than an attempt to equalise compute.

3.4 Sequence length and BPC

Within every language, the tokenizer with fewer tokens per 100 characters also has lower BPC. This association is consistent across the three languages, but it should not be interpreted as proof that shorter sequences directly cause lower BPC. Sequence length also changes context coverage and the number of prediction steps.

3.5 Vocabulary allocation

The shared BPE vocabulary is not divided evenly between the languages. Chinese receives about half of the vocabulary entries, while English and Turkish share a much smaller portion. However,

Tokenizer	English	Turkish	Chinese
Character	2.019	2.084	6.245
BPE-2k	1.987	2.033	6.575
BPE-10k	1.972	2.033	6.186

Table 4: Test bits per character (BPC). Lower is better.

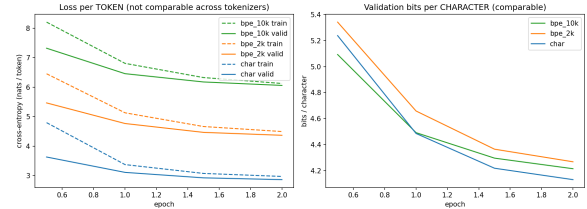


Figure 3: Validation learning curves for the three tokenizers.

Chinese still needs many more distinct tokens to cover its text.

Chinese receives 54% of the 2k vocabulary and 49% of the 10k vocabulary. English and Turkish share only 16% to 18% of the entries, but those short Latin pieces cover a large fraction of their text.

3.6 Error analysis

The sentence-level analysis shows that BPE-2k can be particularly expensive for Chinese sentences containing rare characters.

What the experiment shows: In sentences full of rare Chinese characters, BPE-2k splits most characters into byte pieces (53 tokens for 33 characters) and is much more expensive than the character model (+3.9 bits per character in the first sentence). BPE-10k is better but still above the character model in that sentence. For English and Turkish, BPE-10k wins on sentences made of common words that became single tokens (*·feet*, *·cm*, *·eski*, *·adı*, *·köy*), and BPE-2k wins on sentences with names and unusual strings (*Merrick*, *Cos Dashit*).

My interpretation (not tested): Rare characters are expensive under BPE-2k because the model has to predict several byte tokens for one character, and those byte sequences are seen rarely. Under BPE-10k, common words become cheap, but a name or an unusual string is cut into pieces that are themselves rare tokens, and the larger output layer spreads the probability over more candidates. I did not run controlled tests for these explanations.

Two group comparisons: Chinese sentences with digits are much cheaper than those without

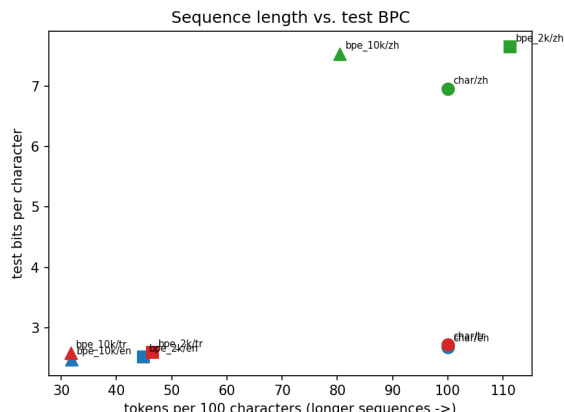


Figure 4: Sequence length and BPC by tokenizer and language.

Language	BPE-2k	BPE-10k
English	15%	18%
Turkish	16%	18%
Chinese	54%	49%

Table 5: Approximate share of vocabulary entries associated with each language.

(BPE-10k 5.47 vs. 6.77; 181 vs. 319 sentences), probably because dates and numbers are very predictable. For Turkish, sentences containing a word of 12 or more characters are *cheaper* (BPE-10k 2.004 vs. 2.084; 301 vs. 199 sentences), the opposite of English (2.065 vs. 1.944; 98 vs. 402). This also appears under the character model (2.202 vs. 2.297), so it is a property of the text, not of the tokenizer. A possible reason is that long Turkish words are chains of suffixes whose later characters are predictable, but this is only a guess.

4 Conclusion

An expectation that was supported. Chinese needs the most tokens, and a larger BPE vocabulary compresses all languages more and, in this experiment, gives better language models: BPE-10k has the lowest test BPC in English (1.972), Turkish (2.033) and Chinese (6.186).

An expectation that was not supported, and a surprise. I expected BPE-2k to be the best choice. It is not: overall it only ties with the character model (3.593 vs. 3.589), because it improves English and Turkish but is clearly worse for Chinese (6.575 vs. 6.245). I was also surprised that Turkish and English compress equally well at 10k but still differ in BPC.

What the experiment shows about a shared tokenizer. In a shared vocabulary, Chinese takes

about half of the entries but gets little compression from them, while English and Turkish share only 16% to 18% of the entries, mostly short Latin pieces. A shared tokenizer therefore works well for the two Latin-script languages and is least efficient for Chinese, and vocabulary size matters much more for Chinese than for the others.

Which approach I would choose. For this multilingual setting I would use the BPE with 10,000 tokens, because it has the best test BPC in every language, beats the character model although it makes fewer updates, and is about the same size as the character model (6.77M vs. 6.48M parameters). Chinese is still its weak spot (7.6% of the Chinese tokens are character fragments and the gain over characters is only 0.9%), so with more time I would first try a larger vocabulary or more Chinese-specific merges, and several seeds.

Limitations

- **One training run per model** (one seed). Small differences, such as character vs. BPE-2k overall (0.004) or the Turkish vs. English gaps (0.03 to 0.06), may be noise.
- **Unequal conditions:** equal epochs give unequal numbers of updates and compute, and the models differ in size (BPE-2k has 59% fewer parameters).
- **Evaluation windows:** I evaluate in consecutive windows of 256 tokens without carrying context across windows. Models that cover less text per window (character, BPE-2k) have more window starts, which may disadvantage them slightly.
- **Character model and unknowns:** `<unk>` never occurs in the training text, so unseen test characters are very expensive for it. They are rare (at most 0.07% of the tokens), so the effect should be small, and it does not apply to BPE.
- **Corpora:** English is Simple English Wikipedia, and the Chinese data mixes traditional and simplified characters.
- **Small models and short training:** the conclusions apply to this scale only.
- **BPC and languages:** BPC values cannot be compared between languages.

Sentence	Character	BPE-2k	BPE-10k
Rare Chinese characters	6.25	10.15	7.20
Common Chinese words	6.33	6.45	5.87
English names	2.31	2.14	2.44
Turkish unusual strings	2.88	2.55	3.08

Table 6: BPC of representative test sentences under the three models.

A Example Tokenizations

One validation sentence per language under the three tokenizers (| separates tokens, · is a space, ◇ is half a character). The character tokenizer gives one token per character: 65 tokens for the English sentence, 68 for the Turkish one and 46 for the Chinese one.

English (BPE-2k: 28 tokens, BPE-10k: 21): *He was First Deputy Chairman of Ways and Means from 2010 to 2013.*

BPE-2k:

He|·was|·F|ir|st|·D|ep|ut|y|·Ch|a|ir|man|·of|·W|ay|s|·and|·M|e|ans|·from|·20|10|·to|·20|13|·

BPE-10k:

He|·was|·F|irst|·Dep|ut|y|·Ch|air|man|·of|·W|ays|·and|·Me|ans|·from|·2010|·to|·2013|·

Turkish (BPE-2k: 25 tokens, BPE-10k: 15): *Se-
lim'in hükümlanlığı sırasında önemli bir değişiklik
daha olmuştur.*

BPE-2k:

S|el|im|'·in|·h|ük|üm|ran|lı|ğ|ı|·sır|asında|·ön|em|li|·bir|·değ|iş|ik|lik|·daha|·olm|uştur|·

BPE-10k:

S|elim|'·in|·hüküm|ran|lığı|·sırasında|·önemli|·bir|·değişik|lik|·daha|·olmuştur|·

Chinese (BPE-2k: 48 tokens, BPE-10k: 33): 列
車運用方面，初期服務以 10 輛編組為主，完全
取代 103 系後才大幅增加以 15 輛編組服務的
班次。

BPE-2k: 列|車|運|用|方|面|，

|初|期|服|務|以|10|◇|◇|◇|◇|組|為|主|，

|完|全|取|代|10|3|系|後|才|大|◇|◇|◇|◇|增|加|以|15|◇|◇|◇|◇|組|服|務|的|班|次|。

BPE-10k: 列車|運|用|方面|，|初期|服

務|以|10|輛|編|組|為主|，|完全|取

代|10|3|系|後|才|大幅|增

加|以|15|輛|編|組|服務|的|班|次|。